

Dharanesh Palaniappan

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Research Profile

Physics graduate with research experience in observational and data-driven astronomy, including systematic analysis of solar features using SDO and OMNI datasets. Experienced in Python-based data analysis, research workflows, and telescope instrumentation. Research interests include solar physics, heliophysics, and astronomical data analysis.

RESEARCH EXPERIENCE

Systematic Analysis of Solar Active Region NOAA AR 12781 (Manuscript in progress)

- **Aim:** Investigated the relationship between magnetic flux evolution and coronal brightness using multi-wavelength SDO (AIA/HMI) observations.
- **Methodology:** Developed a SunPy-based analysis pipeline, applied manual contour tracking for spatial consistency and incorporated a 3-day temporal offset to compare solar observations with OMNI solar wind measurements.
- **Result:** Confirmed an inverse correlation between magnetic flux and coronal intensity during the active region decay phase.

Morphological Analysis of a Galaxy (IUCAA Workshop) | *Nirmala College*

- Analyzed HST FITS imaging using SAOImage DS9 to characterize galaxy morphology and construct a growth curve.
- Derived an angular radius of 2.4 arcseconds and physical radius of 15.3 kpc from background-subtracted photometry and redshift data.
- Calculated photometric properties: Apparent Magnitude ≈ 21.06 and Absolute Magnitude ≈ -21.39 , suggesting a luminosity slightly brighter than the Milky Way.

Comprehensive Analysis of 501 Stars and Their Exoplanetary Systems

- Conducted a systematic analysis of 501 stars selected from the MAST database to search for exoplanet transit signals and cross-match candidates with confirmed host stars.
- Utilized the Lightkurve Python package to process and analyze stellar light curves obtained from Kepler and TESS observations.
- Extracted stellar and planetary parameters from the NASA Exoplanet Archive and performed statistical comparisons between detected signals and confirmed exoplanet systems.

PREPRINT & PUBLICATIONS

Preliminary Analysis of Asteroid Candidate (IASC Campaign) | *Pan-STARRS Data*

- Identified a moving candidate from Pan-STARRS 1 survey data and classified it as a probable Main Belt Asteroid based on its angular velocity (~ 20.6 arcsec/hr).
- Performed Discrete Fourier Transform (DFT) and Signal-to-Noise Ratio (SNR) analysis to distinguish the astronomical signal from instrumental noise.
- Cross-verified the object using the Minor Planet Center (MPC) database, confirming it as the previously designated MBA 2021 CP66.

EDUCATION

Madras Christian College (MCC) | Chennai, India

Jun 2025 – Present

Master of Science (M.Sc.) in Physics | Current GPA: 7.65/10

Madras Christian College (MCC) | Chennai, India

Jul 2022 – Apr 2025

Bachelor of Science (B.Sc.) in Physics | CGPA: 8.06/10

TECHNICAL SKILLS

Data Analysis & Visualization: Python - Sunpy & Lightkurve, Pyspedas, OriginPro, SAOImage DS9, Astrometrica

Laboratory & Observation: Celestron software, Telescope Handling, Electronics

SUMMER SCHOOLS & WORKSHOPS

BRICS Data Analytics Course

- Completed 10-Week BRICS Data Analytics Program: Built skills in Python programming, data structures, and statistical analysis
- Optimized SDO AIA/HMI Data Pipeline: Implemented advanced data collection and retrieval methods for solar datasets to improve extraction efficiency for the BRICS capstone project

The South Asian Astrophysics Summer School

- Analyzed real-time solar data using both remote and in situ measurement techniques.
- This included working with data from the Parker Solar Probe, HMI, and AIA instruments, which enhanced my practical skills in solar observation and advanced data analysis.

Intro2Astro

- * Participated in the Intro2Astro summer workshop, which is designed for undergraduate students.
- * In the summer workshop, we learned the basics of Python, data analysis with real astronomical data, and future career opportunities.

Sagan Summer Workshop

- * Engaged in specialized training on exoplanet detection and characterization techniques, with emphasis on light curve analysis, transit methods, and statistical tools for exoplanetary science.

SCIENTIFIC OBSERVATIONAL EXPERIENCE

Citizen Scientist | *International Asteroid Search Campaign*

- * Analyzed astronomical images via Astrometrica to detect asteroids for NASA-affiliated citizen science initiatives.
- * Refined practical skills in astrometry, data analysis, and data reduction.

Secretary | *B.M. Arthur Astro-Ephemeris Club (MCC)*

- * Organized and led campus astronomy activities, stargazing workshops, and celestial-event viewings.
- * Managed setup, alignment, and operation of Celestron systems and associated control software.

Observatory Volunteer & Assistant | *Devadas Observatory & Star Parties*

- * Assisted observatory staff with telescope handling, routine maintenance, and practical night-sky navigation.
- * Facilitated pointing validation and public astronomy outreach components at stargazing events.

HONORS

International Astronomy and Astrophysics Competition (IAAC)

- * Earned Silver Honor among 14,500+ participants worldwide.

COURSES

Confronting The Big Questions: Highlights of Modern Astronomy | University of Rochester

Astronomy: Exploring Time and Space | University of Arizona

The Science of The Solar System | Caltech

The Diversity of Exoplanets | University of Geneva